

Reminders

Find all links on the course website: cogs9.github.io/fa26



Assignments (find links on course website):

- Reading quiz 1, pre-course survey, syllabus quiz (**DUE: Wed week 2**)
- Reading quiz 2 (**DUE: Wed week 3**) *All due 11:59pm PT*

COGS 9 | Week 2

Data & data types

Brendan Tomoschuk

Today's Learning Goals

1. Define data
2. Distinguish qualitative vs. quantitative data match them to appropriate summaries/visualizations
3. Identify whether data is stored in structured or unstructured formats
4. Choose the appropriate data needed to answer a data science question

What comes to mind when you think of “data”?

What's something in your everyday life that could be considered data?

(~3 min with your neighbor)

Data

- Data is anything that can be **measured** or **observed** about the world
 - If you can enter it into a spreadsheet, take a picture of it, write about it, record it on video or audio— then it is data!
- Data values can be...
 - **Qualitative:** describes categories or qualities
 - Categorical
 - Nominal or *unordered* sets (e.g., colors, blood type, true/false)
 - Ordinal or *ordered* sets (e.g., low, medium, high)
 - Text, audio, images (in raw form)
 - **Quantitative:** a value that can be *objectively* measured or counted
 - Continuous (e.g., height, temperate, population)
 - Discrete (e.g., number of kids, score in a game)

Types of Data

```
graph TD; A[Types of Data] --> B[Quantitative data - numbers]; A --> C[Qualitative data - not numbers]; B --> D[Discrete data]; B --> E[Continuous data]; C --> F[Categorical data]; D --> G["• Number of students<br>• Number of employees<br>• Score of a game"]; E --> H["• Heights of people<br>• Daily temperature<br>• Time needed on a task"]; F --> I["• Hair color<br>• Eye color<br>• Favorite color"];
```

Quantitative data - numbers

Qualitative data - not numbers

Discrete data

Continuous data

Categorical data

- Number of students
- Number of employees
- Score of a game

- Heights of people
- Daily temperature
- Time needed on a task

- Hair color
- Eye color
- Favorite color

Quick check!

Is the data **quantitative** or **qualitative**?

- Number of siblings
- Favorite music genre
- Height
- ZIP code
- Eye color
- Temperature
- Student ID number
- Country of birth
- Shoe size
- Letter grade (A–F)
- GPA
- “Agree/Neutral/Disagree” survey response

Quick check!

Is the data **quantitative** or **qualitative**?

- Number of siblings
- Favorite music genre
- Height
- ZIP code
- Eye color
- Temperature
- Student ID number
- Country of birth
- Shoe size
- Letter grade (A–F)
- GPA
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Data should be visualized according to its type

Quantitative

- Histogram
- Box plot / violin plot
- Scatter plot
- Line chart (over time)

Qualitative

- Bar chart
- Grouped/stacked bar chart
- Pie chart (for few categories)

Webclicker

Sign in with your UCSD email!

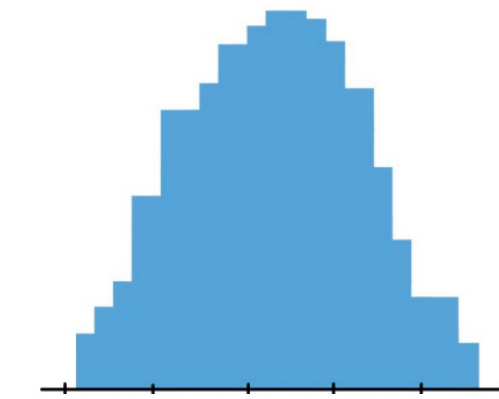
webclicker.web.app



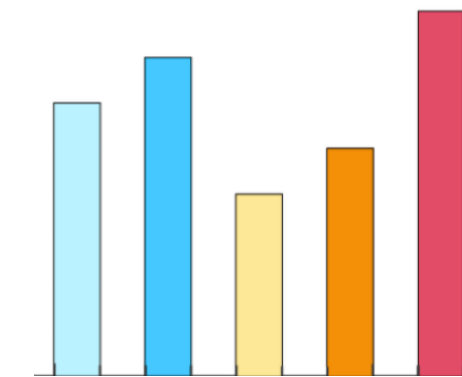
course code:
WSTAPX

What is an appropriate way to summarize height information from 100 people?

(A) Histogram



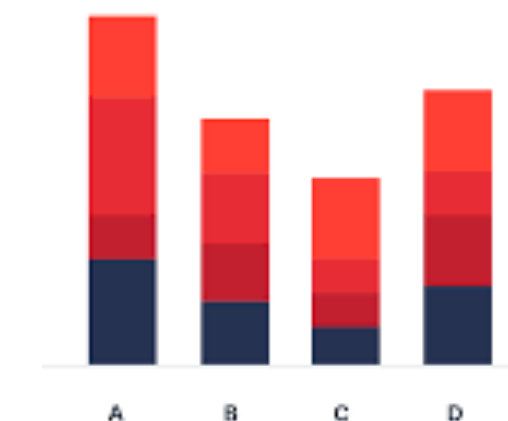
(B) Bar chart



(C) Pie chart



(D) Stacked bar chart



(E) I'm not sure / I'm confused

Webclicker

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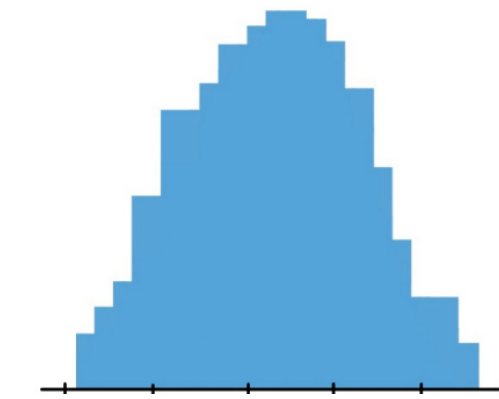
webclicker.web.app



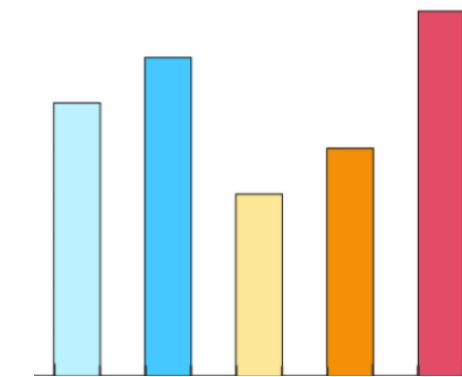
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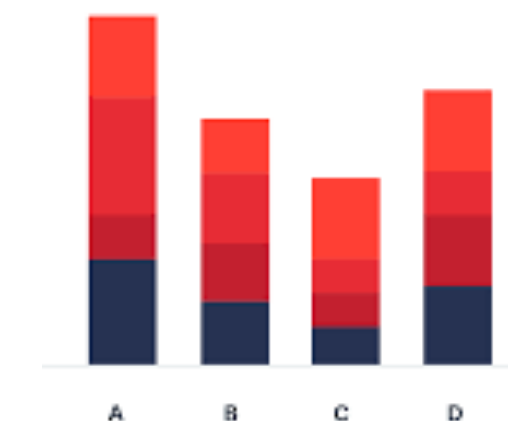
(B) Bar chart



(C) Pie chart



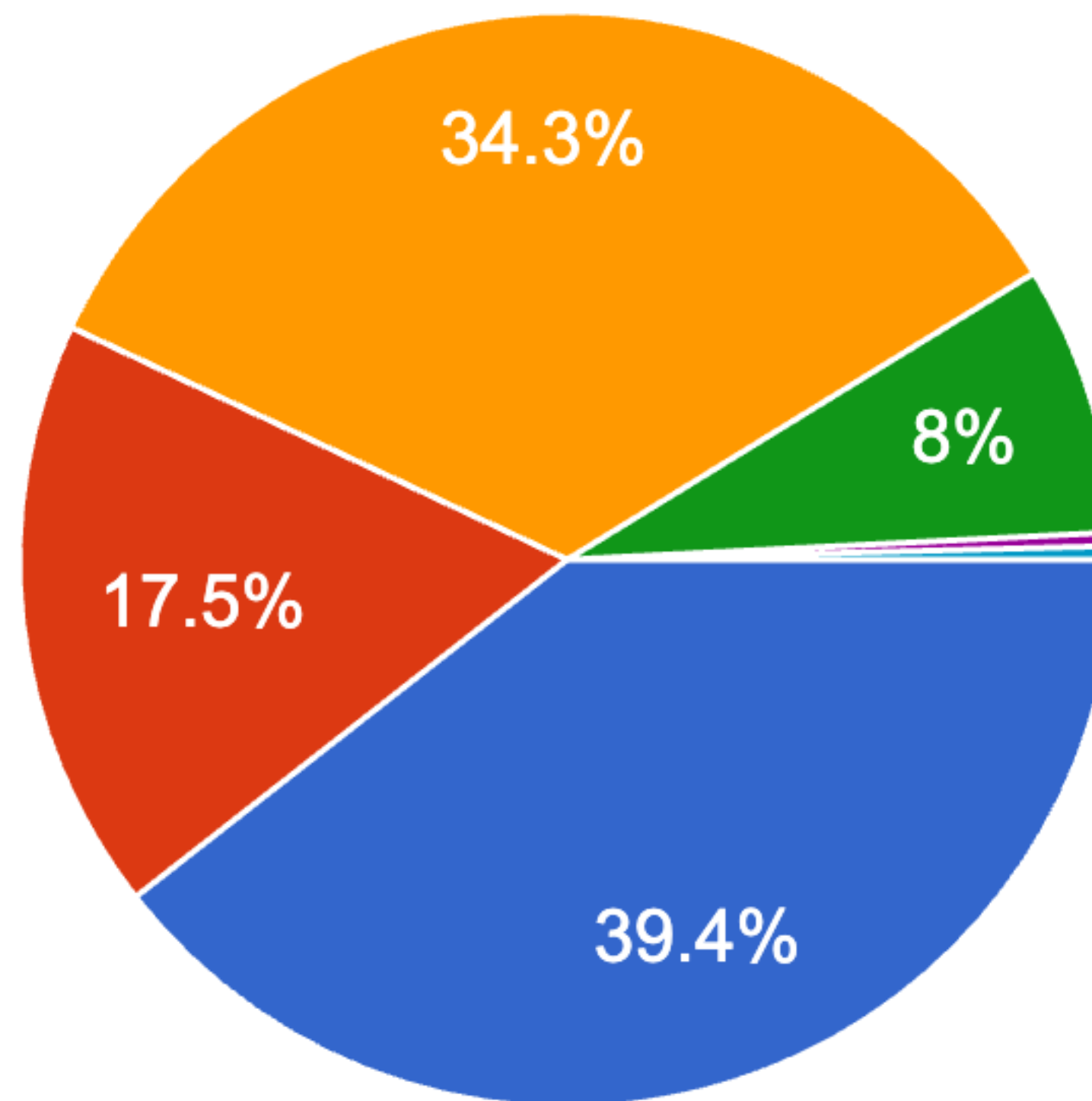
(D) Stacked bar chart



(E) I'm not sure / I'm confused

Year

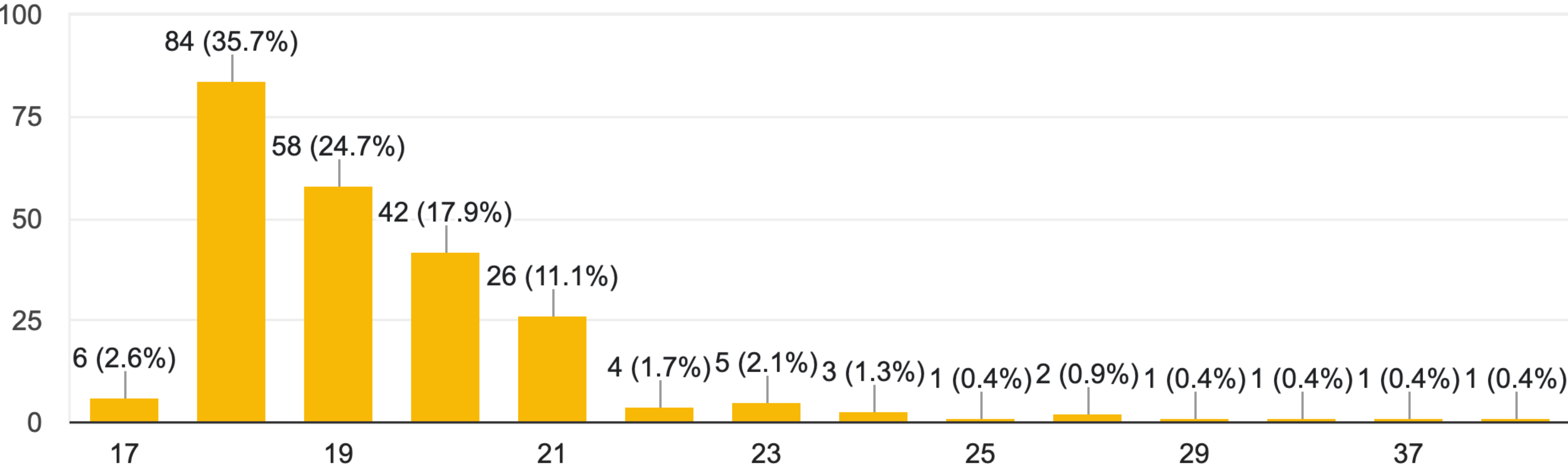
251 responses



- First-year
- Second-year
- Third-year
- Fourth-year
- Fifth-year
- Other

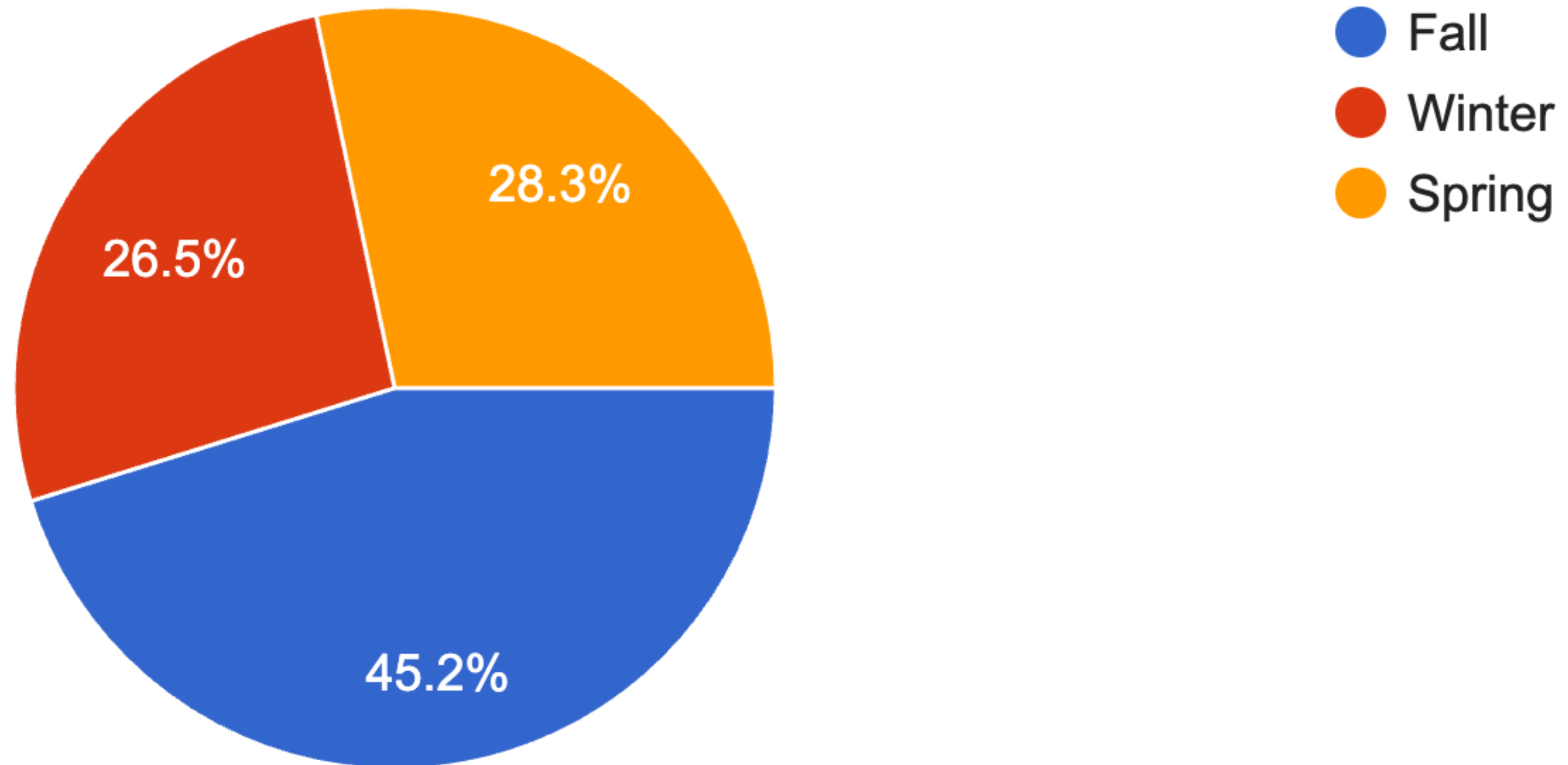
How old are you?

235 responses



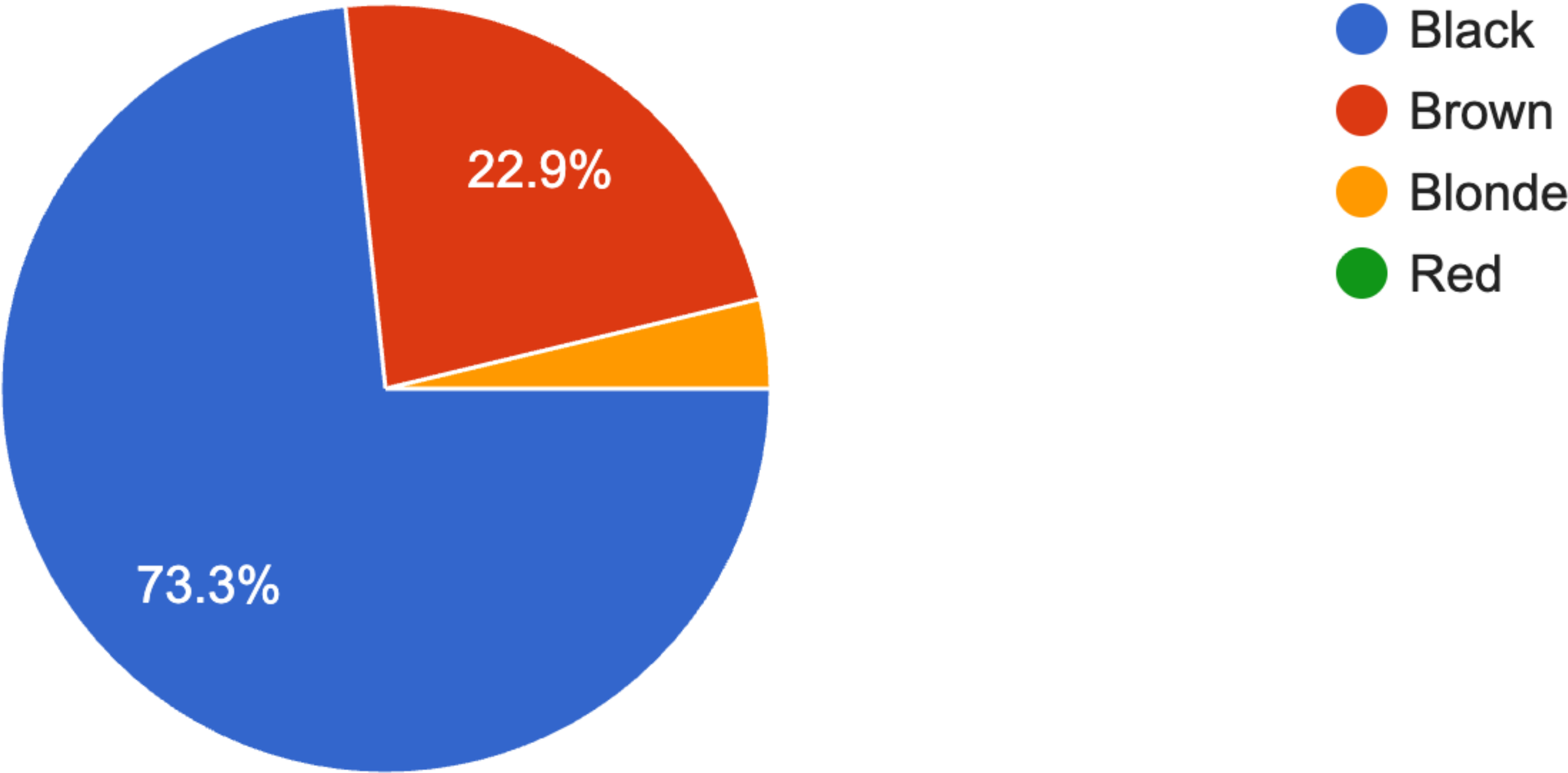
What's your favorite quarter?

219 responses

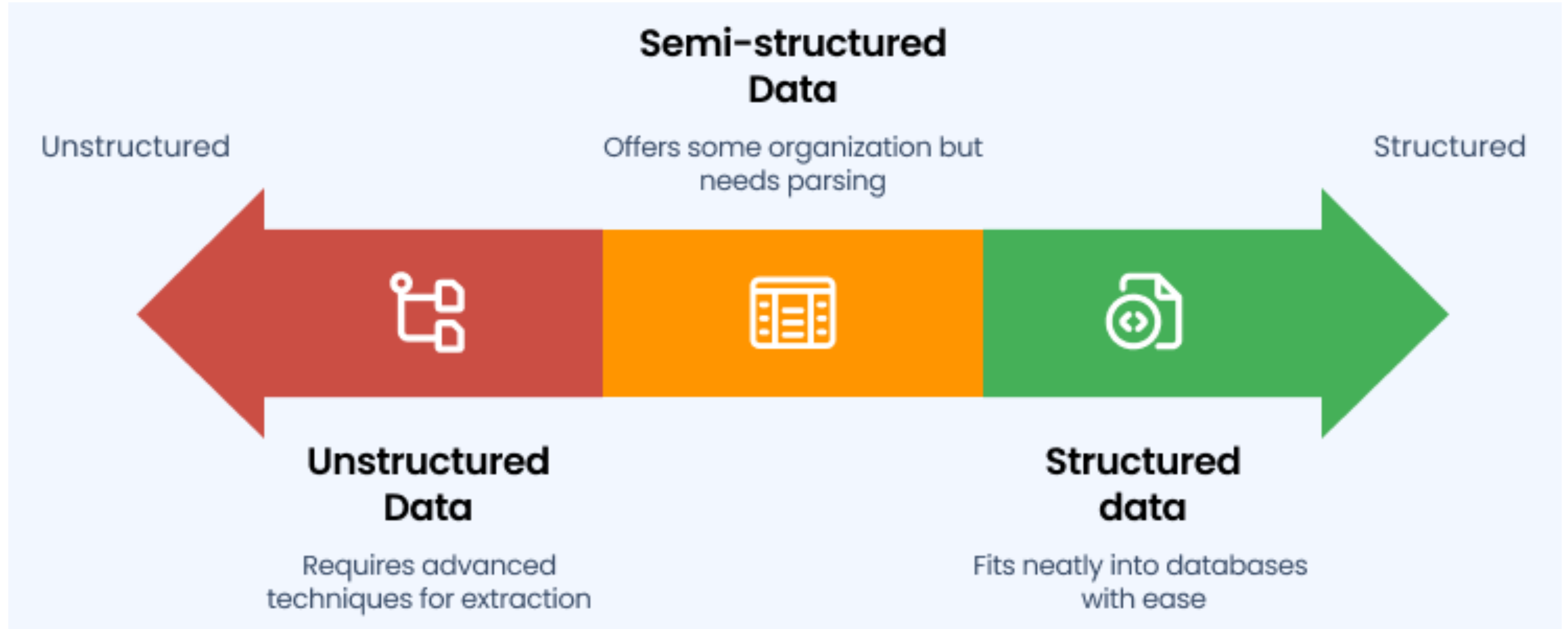


What's your natural hair color?

240 responses



We categorize data storage by how structured it is



Structured data

- Structured data can be organized into an $n \times p$ table of values
- n = the number of observations $n = 12$
- p = the number of features
- Data values can be quantitative or qualitative

College dataset

$$p = 4$$

features OR variables

	Private	Apps	Accept	Enroll
Abilene Christian University	Yes	1660	1232	721
Adelphi University	Yes	2186	1924	512
Adrian College	Yes	1428	1097	336
Agnes Scott College	Yes	417	349	137
Alaska Pacific University	Yes	193	146	55
Albertson College	Yes	587	479	158
Albertus Magnus College	Yes	353	340	103
Albion College	Yes	1899	1720	489
Albright College	Yes	1038	839	227
Alderson-Broaddus College	Yes	582	498	172
Alfred University	Yes	1732	1425	472
Allegheny College	Yes	2652	1900	484

↑
qualitative

← ↑ ↗
quantitative

Structured data

College dataset

- **Private** : Public/private indicator
- **Apps** : Number of applications received
- **Accept** : Number of applicants accepted
- **Enroll** : Number of new students enrolled
- **Top10perc** : New students from top 10 % of high school class
- **Top25perc** : New students from top 25 % of high school class
- **F.Undergrad** : Number of full-time undergraduates
- **P.Undergrad** : Number of part-time undergraduates
- **Outstate** : Out-of-state tuition
- **Room.Board** : Room and board costs
- **Books** : Estimated book costs
- **Personal** : Estimated personal spending
- **PhD** : Percent of faculty with Ph.D.s
- **Terminal** : Percent of faculty with terminal degree
- **S.F.Ratio** : Student/faculty ratio
- **perc.alumni** : Percent of alumni who donate
- **Expend** : Instructional expenditure per student
- **Grad.Rate** : Graduation rate

$n = 12$
observations

$$p = 4$$

features OR variables

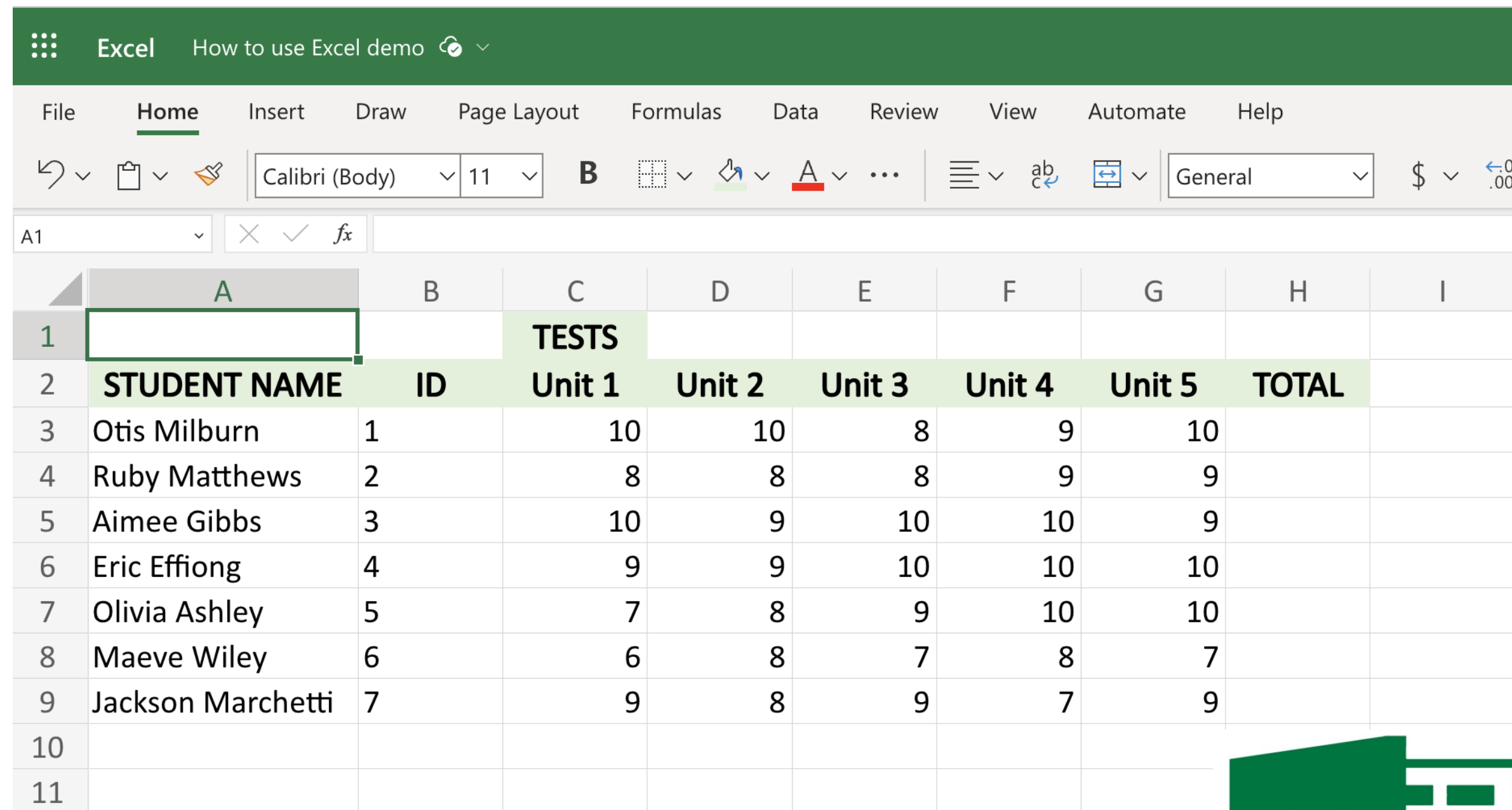
	Private	Apps	Accept	Enroll
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↑ qualitative ↙ quantitative ↑ ↗

Data storage: structured data

Spreadsheets: .csv, Excel (.xls, .xlsx)

Databases: SQL (structured query language)



The image shows a screenshot of an Excel spreadsheet. The spreadsheet has a green header bar with the text 'Excel' and 'How to use Excel demo'. Below the header bar is a ribbon with tabs for File, Home, Insert, Draw, Page Layout, Formulas, Data, Review, View, Automate, and Help. The 'Home' tab is selected. The spreadsheet contains a table with the following data:

	A	B	C	D	E	F	G	H	I
1			TESTS						
2	STUDENT NAME	ID	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	TOTAL	
3	Otis Milburn	1	10	10	8	9	10		
4	Ruby Matthews	2	8	8	8	9	9		
5	Aimee Gibbs	3	10	9	10	10	9		
6	Eric Effiong	4	9	9	10	10	10		
7	Olivia Ashley	5	7	8	9	10	10		
8	Maeve Wiley	6	6	8	7	8	7		
9	Jackson Marchetti	7	9	8	9	7	9		
10									
11									



```
1
2 SELECT Customer.CustomerInfList_FirstName
3 AS "First Name",
4     Customer.CustomerInf_LastName
5 AS "Last Name",
6     CustomerOrders.Amout
7 AS "Order Amount"
8 FROM dbo.Customer_InformationListForSale
9 AS Customer
10 INNER JOIN
11     dbo.OrderTransaction_InformationListForSale
12 AS CustomerOrders
13 ON Customer.ID = CustomerOrders.Customer_InformationListForSale_CustID;
```

Column Alias

Table Alias

Table Alias Usage For Join



Data storage: structured data

SQL is an example of a **relational database**

- Software tool that enables you to **relate tables of data together** and **perform actions** on that data
- Can ask questions (**query**), add, update, or delete information stored in tables
- Essential for managing and analyzing large datasets

Table: Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

SELECT first_name, last_name
FROM Customers
WHERE country = 'USA' **AND** last_name = 'Doe';

first_name	last_name
John	Doe

Data storage: structured data

SQL JOIN

The JOIN operation is a powerful tool!

- Combines rows from two (or more) tables by matching a related column (like a shared ID)
- Can then view the linked information together in a new table

Way more flexible than a spreadsheet!

Table: Customers

customer_id	first_name
1	John
2	Robert
<u>3</u>	David
4	John
<u>5</u>	Betty

Table: Orders

order_id	amount	customer
1	200	10
2	500	<u>3</u>
3	300	6
4	800	<u>5</u>
5	150	8



customer_id	first_name	amount
3	David	500
5	Betty	800

Database Ecosystem

SQL

Traditional RDBMSs



“Modern” SQL DBs



NoSQL

Document



Graph



Takeaway:

There are so many ways to store data!
Each database is tailored for specific use cases.



Multi-Model



Wide Column



Unstructured data

- No simple way to represent it in a table format!



Text files and documents



Server, website and application logs



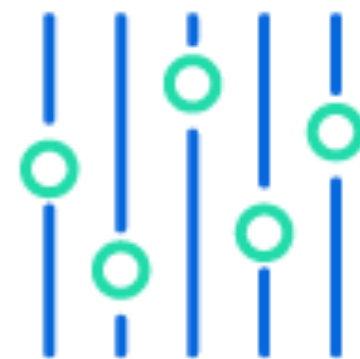
Sensor data



Images



Video files



Audio files



Emails



Social media data

Data storage: unstructured data

- **Text Documents:** .docx, .pdf, .txt, .rtf, .odt
- **Emails & Messaging:** .eml, .msg, instant message logs, chat transcripts
- **Images:** .jpeg, .png, .gif, .tiff
- **Audio:** .mp3, .wav, .aiff, .flac
- **Video:** .mp4, .avi, .mov, .wmv
- **Presentations:** .pptx, .ppt
- **Web Content:** HTML, blog posts, social media feeds



Example: analyzing unstructured data

Sentiment analysis: using **Natural Language Processing (NLP)** to identify and categorize opinions expressed in text

- Is the emotion *positive, negative, neutral*?
- Gauge people's feelings about topics, products, brands, etc. from their written reviews or social media posts

1. Map words to their sentiment

amazing → positive

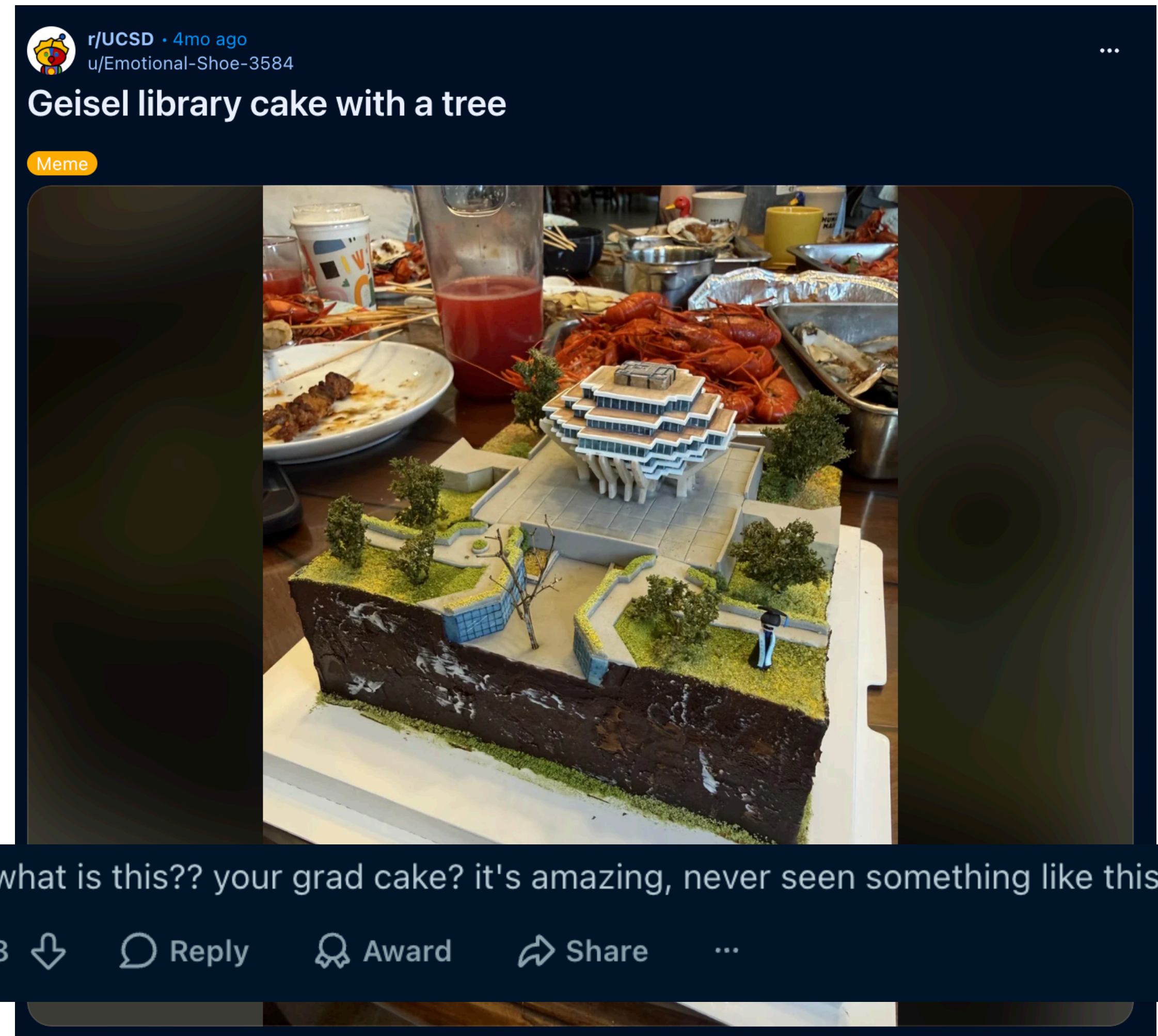
never → slightly negative

2. Convert to a feature score

amazing → +2

never → -1

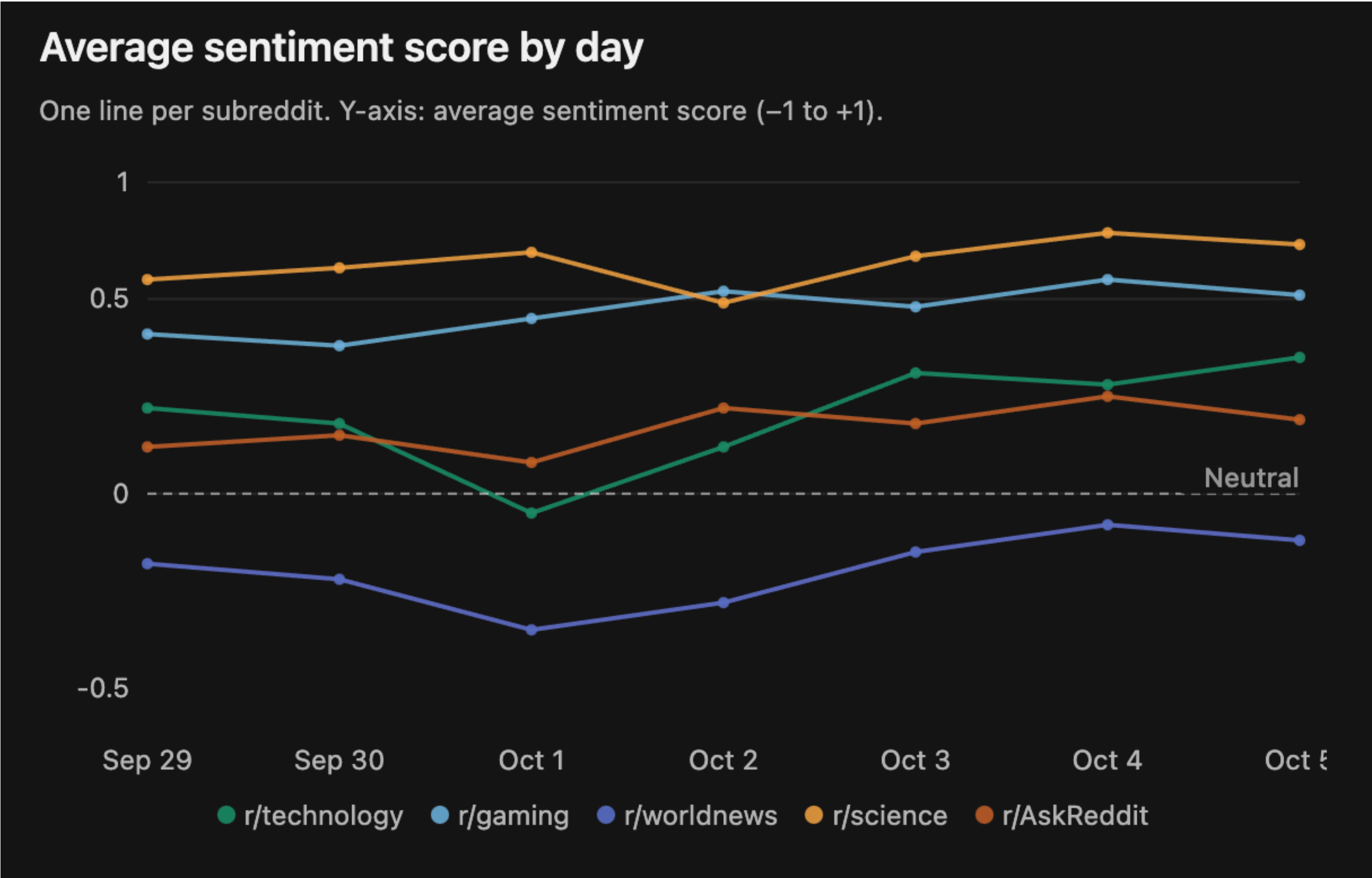
Total: +1 → overall positive!



Example: analyzing structured data

How does each subreddit's sentiment trend over time?

- Calculate average sentiment for all of the posts in a subreddit.
- Plot each day
- Run a *regression* or other statistical model to quantify impact and account for noise



Date	Subreddit	Category	Average sentiment score
Sep 29	r/technology	Tech	0.22
Sep 29	r/gaming	Entertainment	0.41
Sep 29	r/worldnews	News	-0.18
Sep 29	r/science	Science	0.55
Sep 29	r/AskReddit	Discussion	0.12
Sep 30	r/technology	Tech	0.18



Structured Data



Unstructured Data

PDFS

WORD DOCUMENTS

SPREADSHEETS

PRESENTATIONS

SOCIAL MEDIA POSTS

BOOKS

Data storage: *semi*-structured data

Semi-structured: doesn't fit neatly into rows and columns, but has labels & keys that give it organization

- More flexible than databases / spreadsheets, and more organized than unstructured data

```
▼ {
  in_reply_to_status_id : null
  contributors : null
  filter_level : low
  coordinates : null
  id_str : 593891099434983425
  favorited : ☐ false
  created_at : Thu Apr 30 21:34:06 +0000 2015
  favorite_count : 0
  in_reply_to_status_id_str : null
  ▶ retweeted_status : { 24 props }
  in_reply_to_user_id : null
  in_reply_to_screen_name : null
  place : null
  id : 593891099434983425
  retweeted : ☐ false
  truncated : ☐ false
  text : RT @KirkKus: Indirect cost of the UK being in the EU
        is estimated to be costing Britain £170 billion per
        year! #BetterOffOut #UKIP
  in_reply_to_user_id_str : null
  lang : en
  retweet_count : 0
  timestamp_ms : 1430429646725
  user : { 38 props }
```

{.json}

JSON (JavaScript Object Notation)

XML (eXtensible Markup Language)

```
<ROOT>
  <Customers>
    <Customer CustomerName="Arshad Ali" CustomerID="C001">
      <Orders>
        <Order OrderDate="2012-07-04T00:00:00" OrderID="10248">
          <OrderDetail Quantity="5" ProductID="10" />
          <OrderDetail Quantity="12" ProductID="11" />
          <OrderDetail Quantity="10" ProductID="42" />
        </Order>
      </Orders>
      <Address> Address line 1, 2, 3</Address>
    </Customer>
    <Customer CustomerName="Paul Henriot" CustomerID="C002">
      <Orders>
        <Order OrderDate="2011-07-04T00:00:00" OrderID="10245">
          <OrderDetail Quantity="12" ProductID="11" />
          <OrderDetail Quantity="10" ProductID="42" />
        </Order>
      </Orders>
      <Address> Address line 5, 6, 7</Address>
    </Customer>
    <Customer CustomerName="Carlos Gonzlez" CustomerID="C003">
      <Orders>
        <Order OrderDate="2012-08-16T00:00:00" OrderID="10283">
          <OrderDetail Quantity="3" ProductID="72" />
        </Order>
      </Orders>
      <Address> Address line 1, 4, 5</Address>
    </Customer>
  </Customers>
</ROOT>
```

XML

Quick check!

Is the data **structured** or **unstructured**?

- Spreadsheet of students' GPAs and majors
- Audio recordings of meetings
- SQL table of online orders (order_id, item, price, timestamp)
- CSV of daily temperatures by city
- Database of hospital lab results (test name, value, units, date)
- Customer support chat transcripts
- PDF scans of handwritten forms

Quick check!

Is the data **structured** or **unstructured**?

- Spreadsheet of students' GPAs and majors
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- Customer support chat transcripts
- PDF scans of handwritten forms

Discussion (~7 mins)

With a group of 2-3:

Where and when do bike/scooter accidents most likely happen around campus, and what interventions would reduce them?

1. What is the goal(s) of this question?
(Describe, predict, test cause-and-effect, and decide)
2. What data you would need to collect or obtain to answer this question?
3. For each type of data, is it **quantitative** or **qualitative**? Is it **structured** or **unstructured**?

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course code:

WSTAPX

Which of the following is NOT a file format for storing unstructured data?

(A) .pptx

(B) .jpg

(C) .json

(D) .pdf

(E) I'm not sure / I'm confused

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WSTAPX

Which of the following is NOT a file format for storing unstructured data?

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Takeaways

- Data can be classified into **quantitative** or **qualitative** data types
 - Different data types require different summaries / visualizations
- Data can be stored in a variety of formats ranging from **unstructured** to **structured**
- Let your data science question **guide your choice of data!**